

Solar system

Have you ever wondered what a miniature arrangement of our solar system would look like in your bedroom? You might find that you're surprised by the size of the planets, as well as the actual distances between them.

In this chapter, we will take a look at the geometrical arrangement of our solar system, which should give you a better idea of its true scale.

It's next to impossible for us to picture the actual size of the Earth. For this reason, we need to scale it to get an idea of its size.

Let's assume the following scale convention: 1 centimeter (cm) : 1.274.200.000 cm (this means that every centimeter in your room equals 1.274.200.000 centimeters or 12.742 kilometers in the real world). Based on this scale, the diameter of the Earth would be 1 cm.

The most important things to know are the sizes of the planets and the sun, and the distances between them. The gaps between the objects within our solar system are so enormous that it would be incredibly tedious to use centimeters, meters or kilometers; there would be so many zeros that our sense of the ratios would fall apart. So instead, we will use Astronomical Units (AU) to define distance in the solar system. 1 AU is the distance from the sun to Earth, and it equals 149.597.870.700 meters.

$$1 \text{ AU} = 149.597.870.700 \text{ m}$$





This table shows the diameters of the astronomical objects within our solar system. The first column contains the name of the object, the second column shows the actual width, and the third shows one the scaled diameter:

Object's name	Diameter(km)	Scaled diameter (cm)
Sun	1.391.000	109.16
Mercury	4.879	0.38
Venus	12.104	0.94
Earth	12.742	1
Mars	6.779	0.53
Jupiter	139.820	10.9
Saturn	116.460	9.1
Uranus	50.724	3.9
Neptune	49.244	3.8

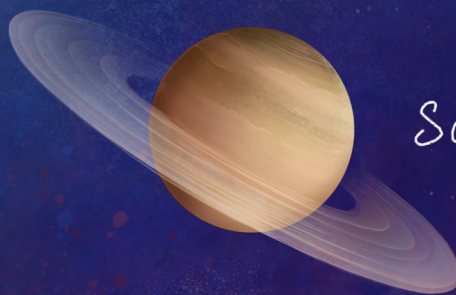
Solar System.



Neptune 8



Uranus 7



Saturn 6



Jupiter 5



Mars 4



Earth 3



Venus 2



Mercury 1

Sun

The next table shows the distances between objects in our solar system. In the following table, the first column shows the name of the object, the second column its actual distance from the sun, and the third column its scaled distance from the sun. Because the numbers are so huge, you would not be able to create a scaled layout of the solar system within your bedroom!

Object's name	Distance from the Sun (AU)	Scaled distance from the Sun (in the METERS)
Sun	0	0
Mercury	0.4	46
Venus	0.7	82
Earth	1	117
Mars	1.5	176
Jupiter	5.2	610
Saturn	9.5	1.115
Uranus	19.2	2.254
Neptune	30.1	3.533

As you can see, the true scale of the solar system is very different from what we encounter in the media. It is clear that the actual proportions are unbelievably uncomfortable for our brains to process. An illustration drawn to scale would have a lot of empty space, and that doesn't make much sense.

Try to think about another astronomical object on your own, using data gathered from the internet. For example: what would be the scaled distance to TRAPPIST-1 - one of many planetary systems which may be able to support life?